

Element H: Prototype Testing and Data Collection Plan

Means of Testing Prototype:

In order to test the prototype, a test plan was developed which would be effective in determining if the prototype meets the stakeholders requirements and is an effective solution to contributing to special needs physical education development.

	Requirement	Testing setup and procedure	Test cases and number of trials	Expected range of results	Pass/fail criteria, goal
1	The prototype must be launch the frisbee a distance of 30 feet	1) Place prototype on the floor or a flat surface 2) Turn prototype on and wait for flywheels to spin at the maximum speed 3) Slowly slide the frisbee up the ramp and make contact with the flywheel 4) Measure distance of where the frisbee landed and record the distance	# of trials: 5	Flywheels may stop spinning at the optimal speed after feeding in the frisbee multiple times resulting in malfunction; the frisbee may start experiencing air resistance which affects its performance leading it to land elsewhere; frisbee may not be able to launch as far due to lack of proper calibration	Pass: The frisbee must land at a distance of 30 feet away in terms of a horizontal distance, the prototype must stay intact through all trials and no malfunctions occur. Fail: Frisbee lands less than 30 feet away from prototype; prototype malfunctions or flywheels stop spinning effectively at any moment. Goal: The prototype must meet the student's needs of launching the frisbee a distance of 30 feet at minimum for it to be

					impactful.
2	Prototype outputs a sound measure of below 70 dBA	1) Place prototype on the floor or a flat surface 2) Turn prototype on and wait for flywheels to spin at the maximum speed 3) Record dbA level using sound level meter 4) Slowly slide the frisbee up the ramp and make contact with the flywheel, while doing so record the dbA level	# of trials: 5	The motor may begin to burn out due to repetitive use and operation; recorded sound measure may exceed or be below the requirement measure	Pass: The prototype outputs a sound measure of below 70 dbA before and during the flywheel is sent through. Fail: The prototype exceeds the required sound measure which requires the use of sound proofing walls to cover the wheels and the motor to limit sound. Goal: Many special needs kids have trouble focusing on a certain task, so the excessive sound may alter their interest with the device, as a result ensuring the prototype is relatively quiet will ensure that the kids receive quality experience.